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Studierichting: Informatica
Oefeningenreeks 4A nr. 14.2

$$I = \int x \ln x dx$$

$$\begin{cases} u = \ln x \\ dv = x dx \end{cases} \Rightarrow \begin{cases} du = \frac{1}{x} dx \\ v = \frac{x^2}{2} \end{cases}$$

$$I = \ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$= \frac{1}{2} x^2 \ln x - \int \frac{x^2}{2x} dx$$

$$= \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x dx$$

$$= \frac{1}{2} x^2 \ln x - \frac{x^2}{4} + c$$

References